Lab Mid

CSE 4308

DATABASE MANAGEMENT SYSTEMS LAB

Prepared by: Samia Nawsheen
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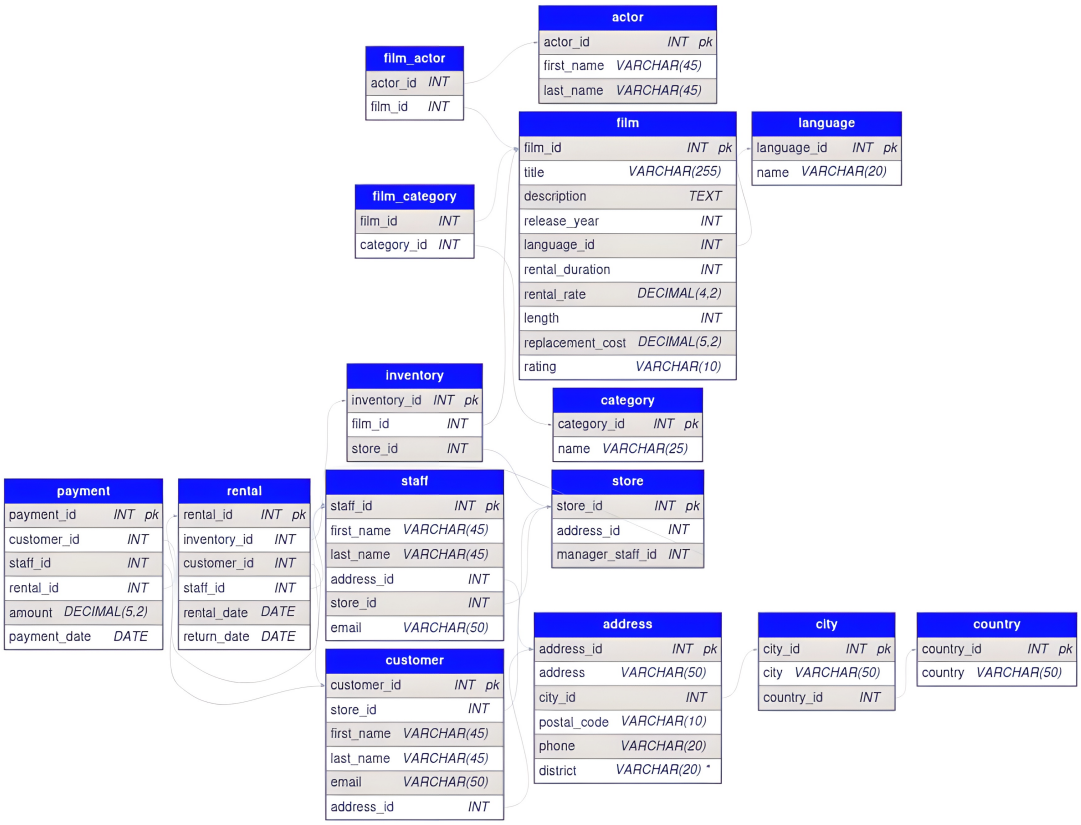
DBMS Lab Mid



1. Display the titles of films that start with a vowel.

```
SELECT title
FROM film
WHERE LOWER(title) LIKE 'a%'
   OR LOWER(title) LIKE 'e%'
   OR LOWER(title) LIKE 'i%'
   OR LOWER(title) LIKE 'o%'
   OR LOWER(title) LIKE 'u%';
```

2. List all distinct film ratings available in the database.

```
SELECT DISTINCT rating
FROM film;
```

3. Display the ID, first and last name for customers who have rented films from every store in the database.

```
SELECT c.customer_id, c.first_name, c.last_name
FROM customer c
JOIN rental r ON c.customer_id = r.customer_id
JOIN inventory i ON r.inventory_id = i.inventory_id
GROUP BY c.customer_id, c.first_name, c.last_name
HAVING COUNT(DISTINCT i.store_id) = (SELECT COUNT(*) FROM store);
```

4. Identify customers who rented films in at least two different months.

```sql
SELECT customer_id
FROM rental
GROUP BY customer_id
HAVING COUNT(DISTINCT EXTRACT(MONTH FROM rental_date)) >= 2;
```

5. Find the total number of rentals handled by each staff member. Display staff name and rental count.

```sql
SELECT s.first_name, s.last_name, COUNT(r.rental_id) AS rental_count
FROM staff s
JOIN rental r ON s.staff_id = r.staff_id
GROUP BY s.staff_id, s.first_name, s.last_name;
```

6. Find films whose rental duration is longer than 5 days.

```sql
SELECT title, rental_duration
FROM film
WHERE rental_duration > 5;
```

7. Show the total number of films available in each category.

```sql
SELECT ca.name AS category, COUNT(fc.film_id) AS film_count
FROM category ca
JOIN film_category fc ON ca.category_id = fc.category_id
GROUP BY ca.category_id, ca.name;
```

8. Find customers whose average rental duration is greater than 3 days.

```sql
SELECT c.customer_id, c.first_name, c.last_name
FROM customer c
JOIN rental r ON c.customer_id = r.customer_id
WHERE r.return_date IS NOT NULL
GROUP BY c.customer_id, c.first_name, c.last_name
HAVING AVG(EXTRACT(DAY FROM (r.return_date - r.rental_date))) > 3;
```

9. Find customers who have rented films from every category available in the database.

```sql
SELECT c.customer_id, c.first_name, c.last_name
FROM customer c
JOIN rental r ON c.customer_id = r.customer_id
JOIN inventory i ON r.inventory_id = i.inventory_id
JOIN film_category fc ON i.film_id = fc.film_id
GROUP BY c.customer_id, c.first_name, c.last_name
HAVING COUNT(DISTINCT fc.category_id) =
       (SELECT COUNT(*) FROM category);
```

10. For each category, list the actor(s) who appeared in the most films in that category.

```sql
SELECT
    c.name AS category_name,
    a.first_name,
    a.last_name,
    COUNT(*) AS film_count
FROM film_category fc
JOIN film_actor fa ON fc.film_id = fa.film_id
JOIN actor a ON fa.actor_id = a.actor_id
JOIN category c ON fc.category_id = c.category_id
GROUP BY c.category_id, c.name, a.actor_id, a.first_name, a.last_name
HAVING COUNT(*) = (
    SELECT MAX(actor_film_count)
    FROM (
        SELECT fa2.actor_id, COUNT(*) AS actor_film_count
        FROM film_category fc2
        JOIN film_actor fa2 ON fc2.film_id = fa2.film_id
        WHERE fc2.category_id = c.category_id
        GROUP BY fa2.actor_id
    ) AS sub
);
```